

JavaScript Arrays & Strings Cheatsheet

Common built-in methods, quick reference

MUT mutates original **PURE** returns new value, original unchanged

Array — Add / Remove

push(...items)	add to end	MUT
pop()	remove & return last	MUT
unshift(...items)	add to start	MUT
shift()	remove & return first	MUT
splice(start,del,...items)	remove/insert at index	MUT
slice(start,end)	extract subarray	PURE
concat(...arrs)	merge arrays	PURE
fill(val,start,end)	fill with value	MUT
copyWithin(t,s,e)	copy part within array	MUT

Array — Search / Test

indexOf(val)	first index or -1
lastIndexOf(val)	last index or -1
includes(val)	true/false membership
find(fn)	first matching element
findIndex(fn)	index of first match
findLast(fn)	last matching element
findLastIndex(fn)	index of last match
some(fn)	true if any pass test
every(fn)	true if all pass test

Array — Transform

map(fn)	new array from fn(el)	PURE
filter(fn)	elements passing test	PURE
reduce(fn,init)	fold to single value	
reduceRight(fn,init)	reduce right-to-left	
flat(depth)	flatten nested arrays	PURE
flatMap(fn)	map then flatten 1 level	PURE
sort(fn)	sort in place	MUT
toSorted(fn)	sorted copy	PURE
reverse()	reverse in place	MUT
toReversed()	reversed copy	PURE
toSpliced(...)	splice on a copy	PURE
with(index,val)	copy with element replaced	PURE

Array — Iterate / Convert

forEach(fn)	run fn on each element
join(sep)	array → string
keys() / values()	iterator over indices / values
entries()	iterator of [index, value]
Array.from(iter, fn)	build array from iterable
Array.of(...items)	build array from arguments
Array.isArray(x)	true if x is an array
at(index)	element at index (supports -1)
length	number of elements (property)

String — Search / Test

includes(sub)	true/false substring check
startsWith(sub)	true if begins with sub
endsWith(sub)	true if ends with sub
indexOf(sub)	first index or -1
lastIndexOf(sub)	last index or -1
search(regex)	index of regex match
match(regex)	array of matches or null
matchAll(regex)	iterator of all matches

String — Extract / Transform

slice(start,end)	substring, allows negatives
substring(start,end)	substring, clamps negatives to 0
at(index)	char at index (supports -1)
charAt(index)	char at index
charCodeAt(i)	UTF-16 code at index
toUpperCase()	convert to upper case
toLowerCase()	convert to lower case
trim()	strip whitespace both ends
trimStart() / trimEnd()	strip whitespace one end
padStart(len,pad)	pad from left to length
padEnd(len,pad)	pad from right to length
repeat(n)	repeat string n times
replace(pat,repl)	replace first match
replaceAll(pat,repl)	replace all matches
concat(...strs)	join strings together

String — Convert

split(sep,limit)	string → array of substrings
String(val)	convert any value to string
toString()	convert to string
String.fromCharCode(...c)	codes → string
localeCompare(str)	compare order: -1, 0, 1
normalize(form)	Unicode normalization
length	number of UTF-16 units (property)

Handy Snippets

[...new Set(arr)]	remove duplicates
arr.reduce((a,b)=>a+b,0)	sum of array
Math.max(...arr)	largest value in array
[...str].reverse().join('')	reverse a string
str.split('').sort().join('')	sort chars alphabetically
Array(n).fill(0)	array of n zeros
Array.from({length:n},(_,i)=>i)	range 0..n-1